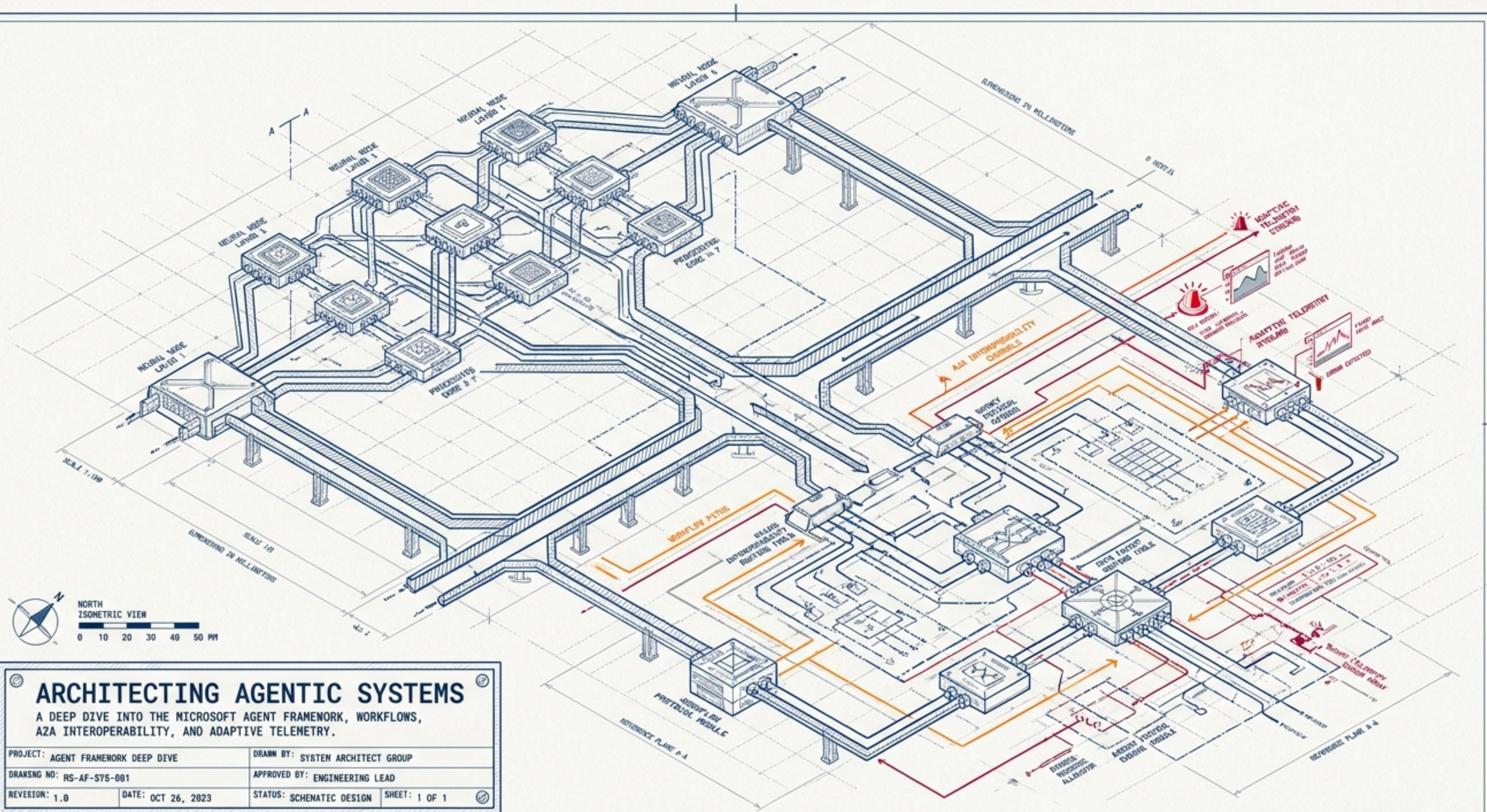


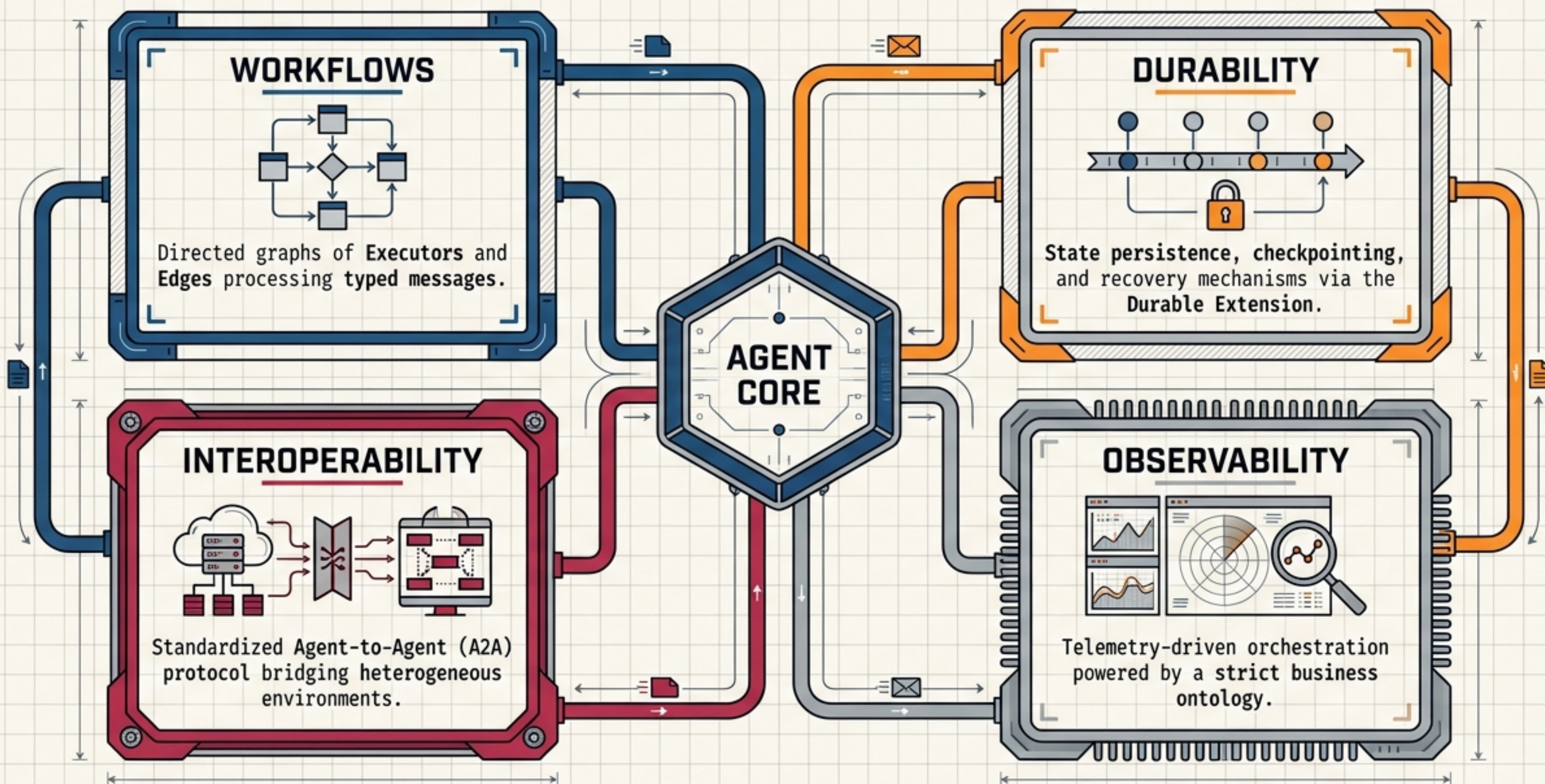


ARCHITECTING AGENTIC SYSTEMS

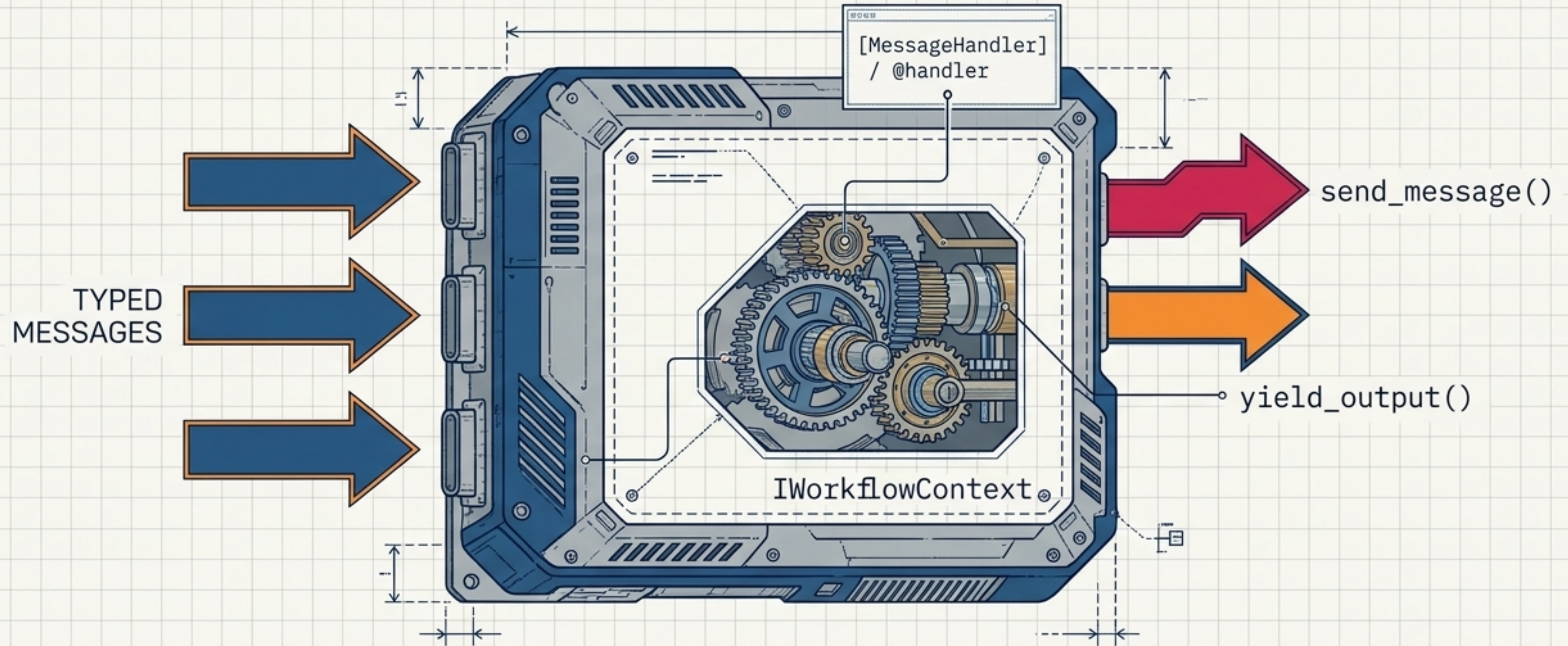
A DEEP DIVE INTO THE MICROSOFT AGENT FRAMEWORK, WORKFLOWS, AZA INTEROPERABILITY, AND ADAPTIVE TELEMETRY.

PROJECT: AGENT FRAMEWORK DEEP DIVE		DRAWN BY: SYSTEM ARCHITECT GROUP	
DRAWING NO: RS-AF-SYS-001		APPROVED BY: ENGINEERING LEAD	
REVISION: 1.0	DATE: OCT 26, 2023	STATUS: SCHEMATIC DESIGN	SHEET: 1 OF 1

FOUR PILLARS OF THE AGENT ECOSYSTEM

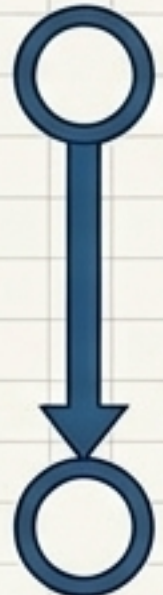
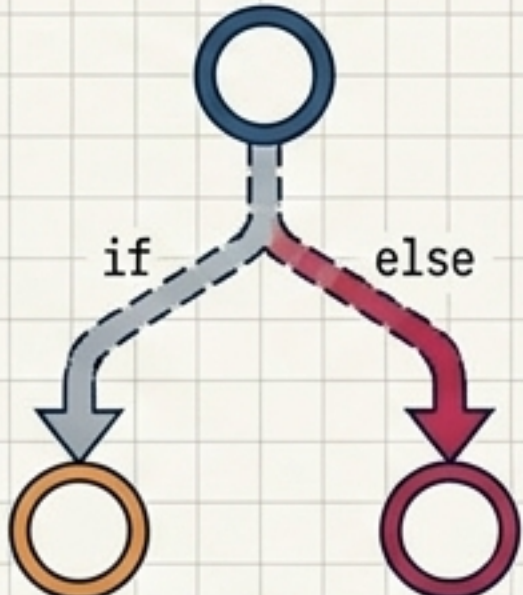
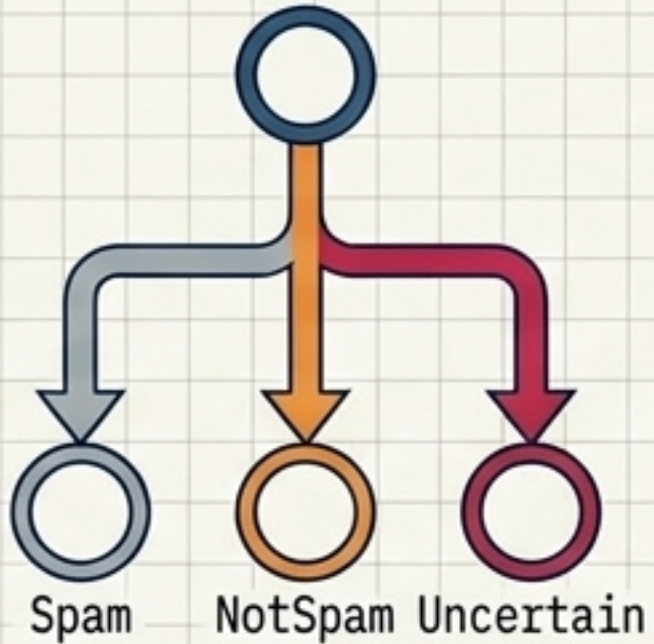
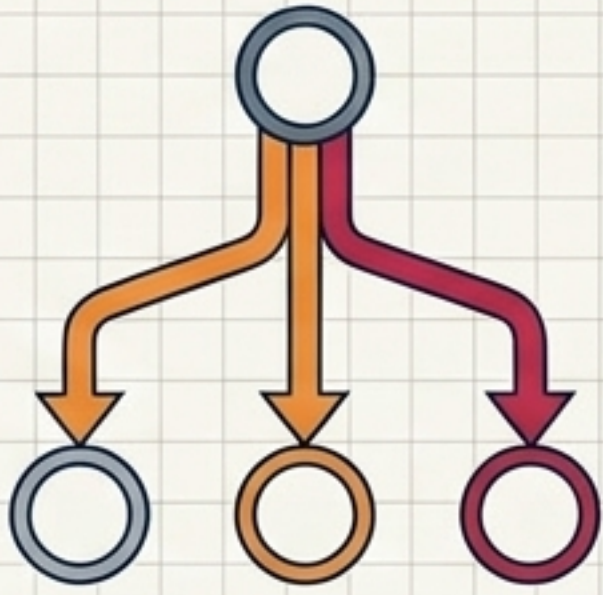


AUTONOMOUS PROCESSING UNITS DRIVEN BY TYPED MESSAGES



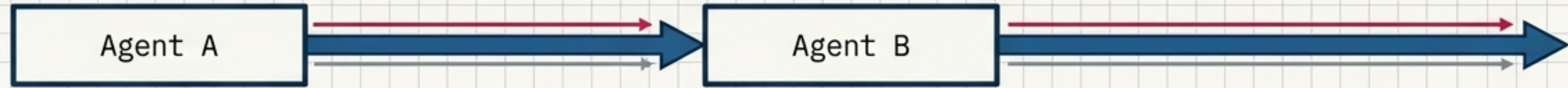
PROJECT: AGENT ECOSYSTEM	
DRAWING NO. AE-002	REV. 1.0

DIRECTING EXECUTION FLOW BETWEEN NODES

DIRECT EDGES	CONDITIONAL EDGES	SWITCH-CASE EDGES	MULTI-SELECTION (FAN-OUT)
			
Linear pipelines (1-to-1 connections). Simplest form of data flow.	Dynamic branching using binary routing (if/else predicates) to evaluate message properties.	Multi-branch routing utilizing a cleaner SwitchBuilder syntax for complex decision trees (e.g., routing as Spam, NotSpam, or Uncertain).	Parallel processing where a single message activates multiple downstream targets simultaneously based on partitioned content constraints.

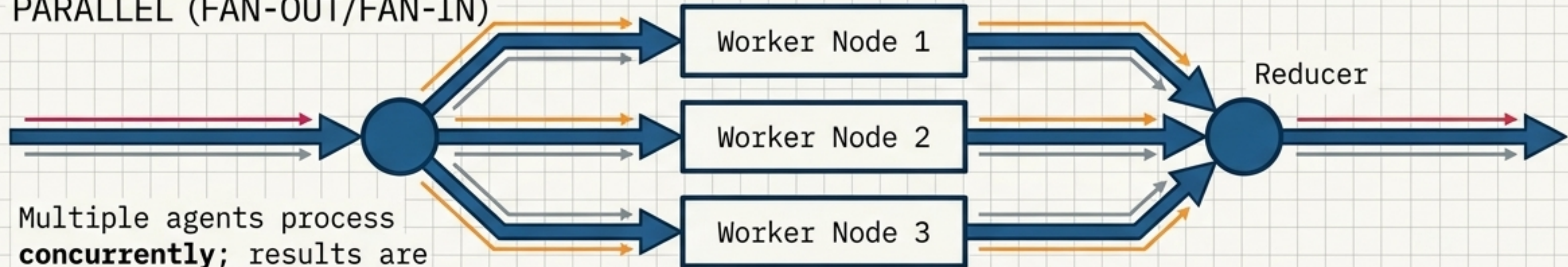
TOPOLOGIES FOR MULTI-AGENT ORCHESTRATION

SEQUENTIAL



Specialized agents pass **refined outputs** forward.

PARALLEL (FAN-OUT/FAN-IN)



Multiple agents process **concurrently**; results are **aggregated**.

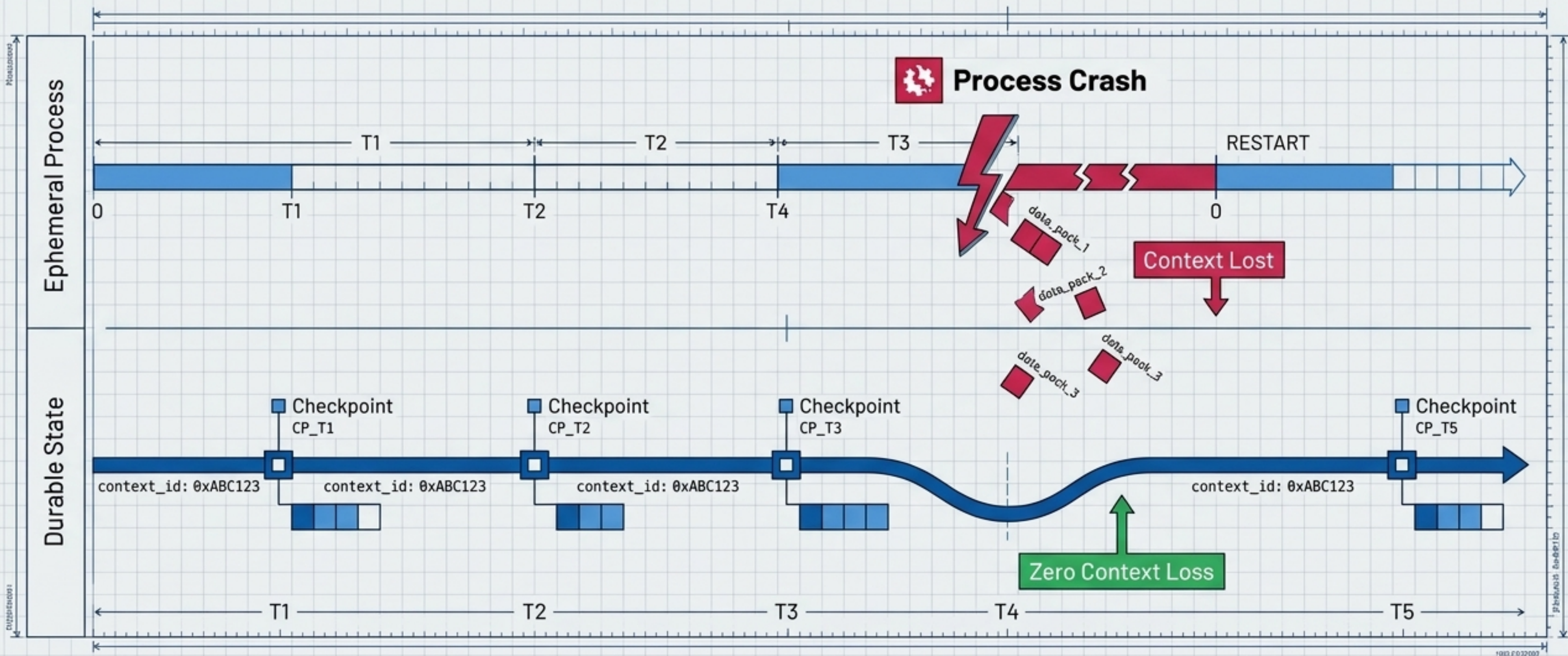
HUMAN-IN-THE-LOOP (HITL)



Execution pauses for **human input**; zero compute consumed while waiting for **external event triggers**.

PROJECT: AGENT ECOSYSTEM	
DRAWING NO: AE-004	REV: 1.0

STATE RESILIENCE THROUGH DURABLE CHECKPOINTING



PROJECT:
AGENT ECOSYSTEM

DRAWING NO.:
AE-005

REV.:
1.0

DEPLOYING THE COMPUTE ENGINE

AGENT ECOSYSTEM - DEPLOYMENT TOPOLOGIES

Azure Functions (Serverless)



- Managed infrastructure
- Built-in HTTP endpoints
- Scale-to-zero capability (**pay-per-invocation**)
- Native Durable Functions integration

Bring-Your-Own-Compute (BYOC)



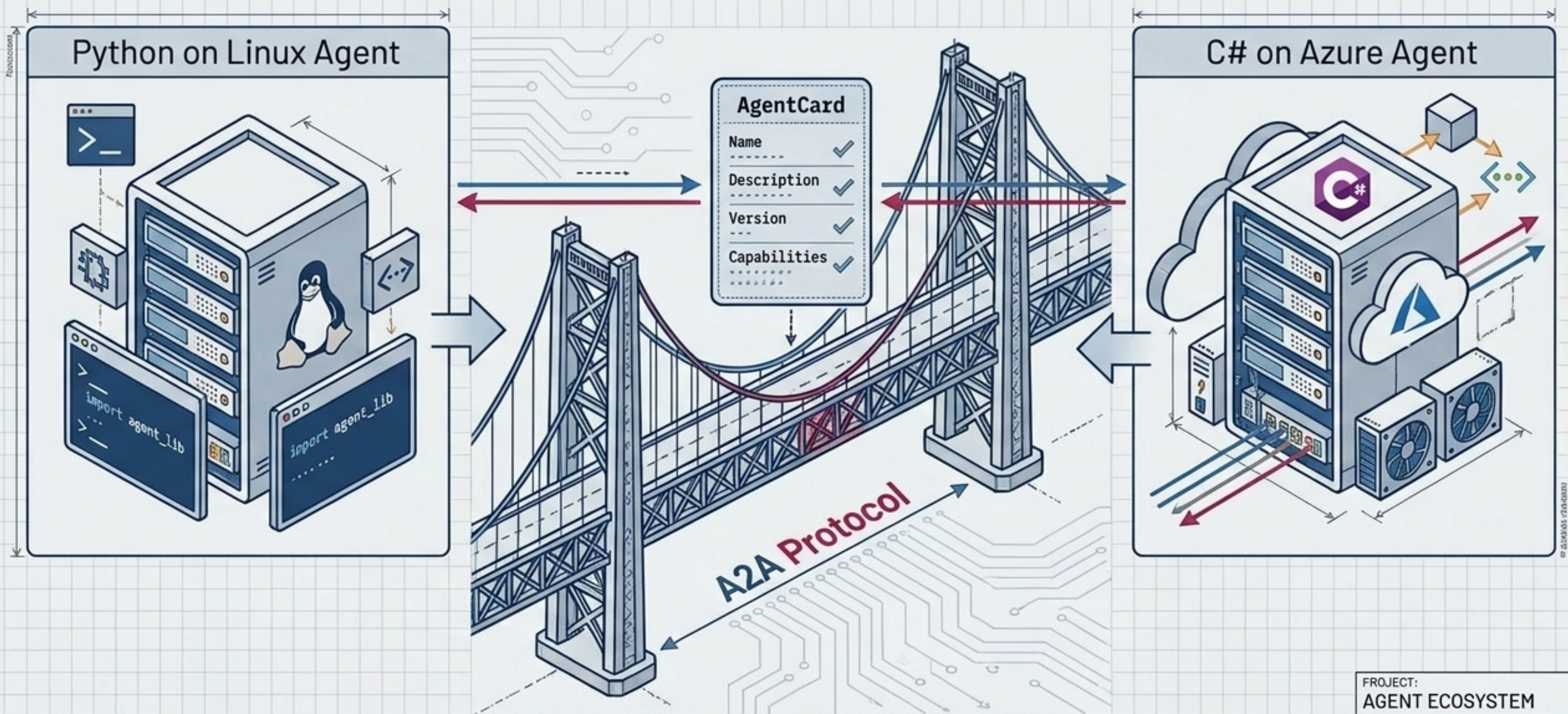
- Self-hosted deployments (**Containers, Kubernetes**)
- Total control over process lifetime and networking
- Integration into existing custom services using **Durable Task Scheduler**

PROJECT:
AGENT ECOSYSTEM

DRAWING NO.:
AE-006

REV.:
1.0

BREAKING SILOS WITH THE AGENT-TO-AGENT PROTOCOL

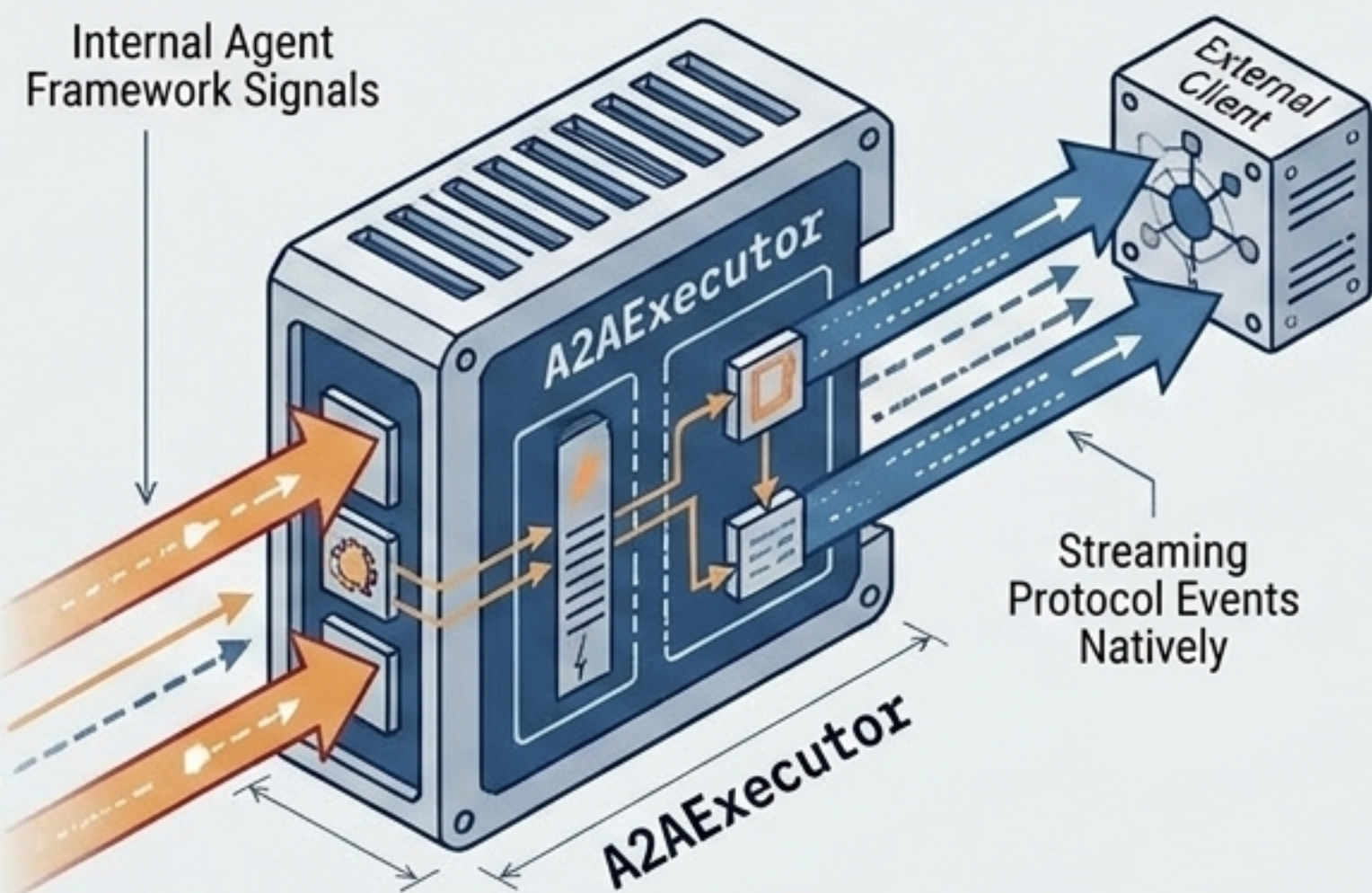


PROJECT: AGENT ECOSYSTEM	
DRAWING NO.: AE-007	REV.: 1.0

EXPOSING AND CONNECTING ACROSS BOUNDARIES

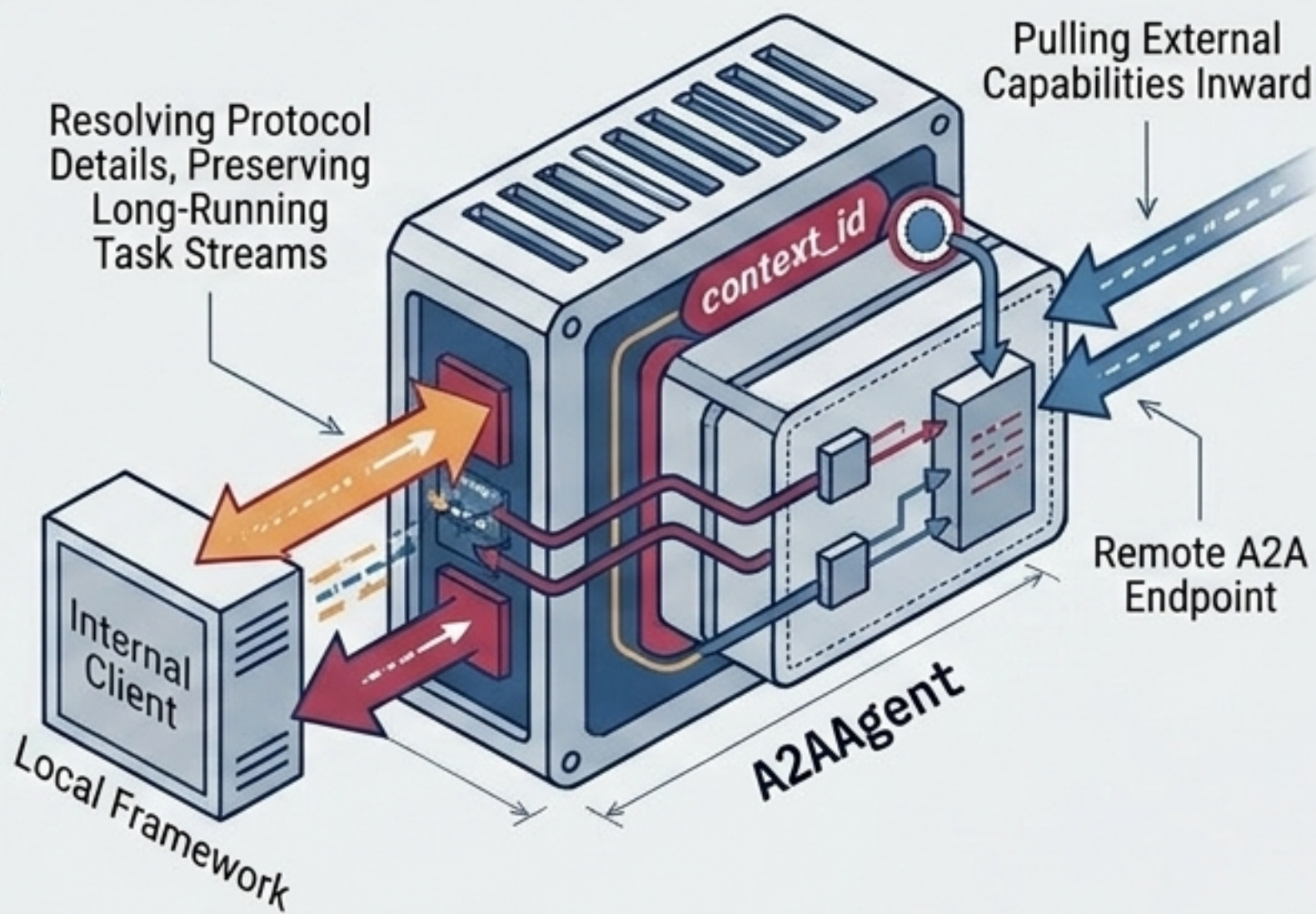
AGENT ECOSYSTEM - A2A LIBRARY

Exposing



Maps internal signals outward, streaming protocol events natively

Connecting

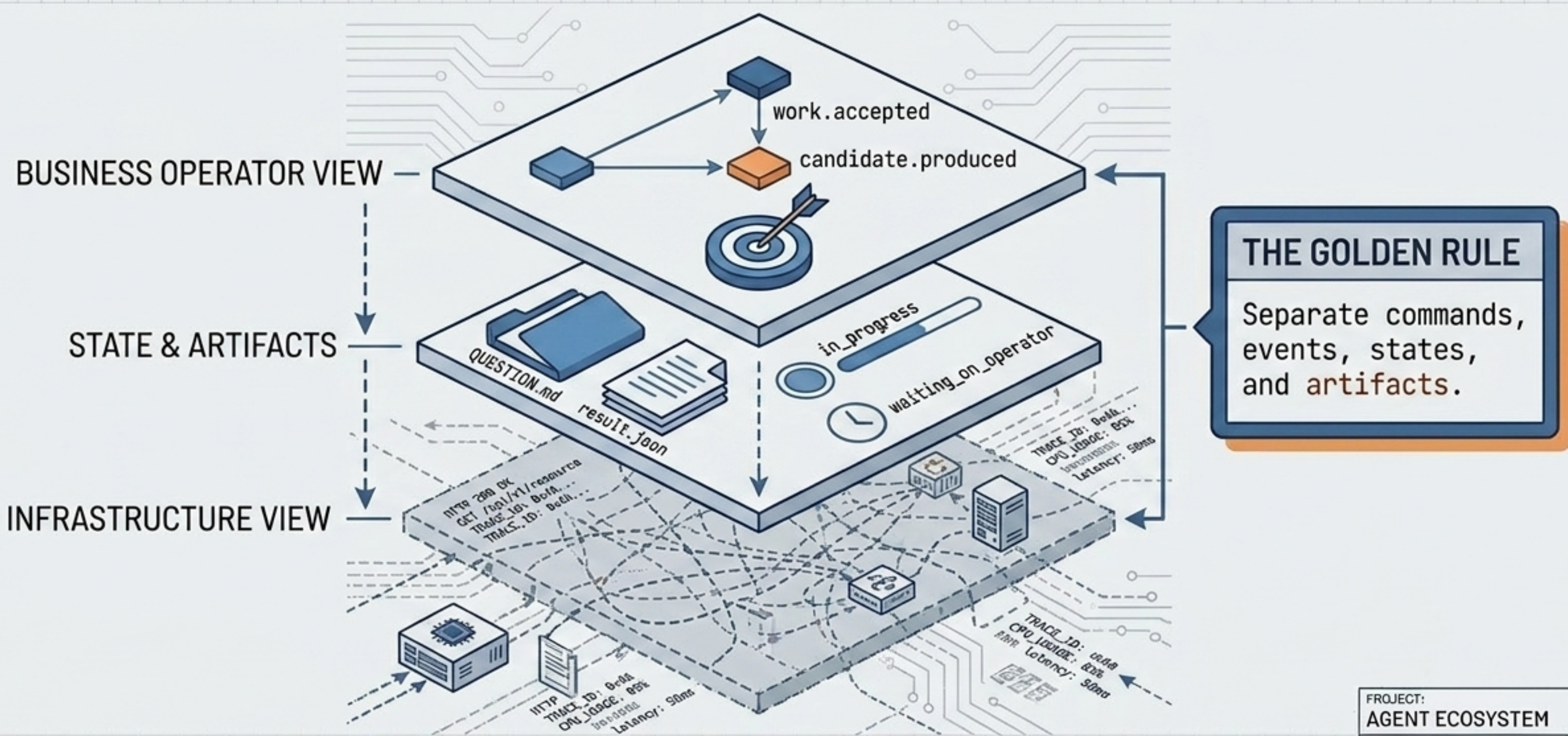


Wraps remote endpoint, resolving protocol details and preserving long-running streams

PROJECT: AGENT ECOSYSTEM	
DRAWING NO.: AE-008	REV.: 1.0

ISOLATING BUSINESS REALITY FROM INFRASTRUCTURE NOISE

OPERATIONAL TRUTH



PROJECT: AGENT ECOSYSTEM	
DRAWING NO.: AE-007	REV.: 1.0

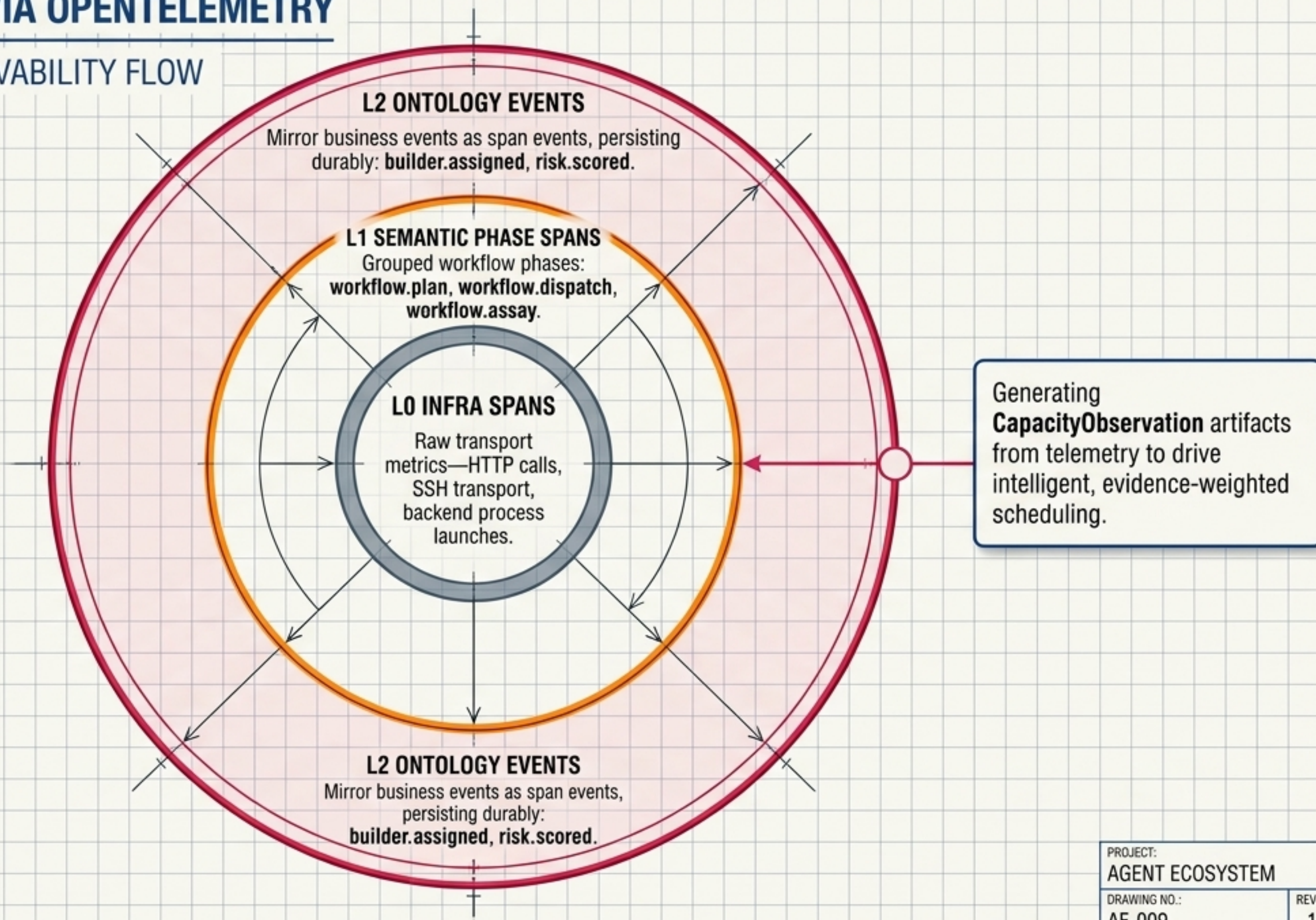
THE INFORMATION REPRESENTATION MATRIX

AGENT ECOSYSTEM – DATA CLASSIFICATION

ONTOLOGY ENTITY	PROJECTED STATE	TELEMETRY ONLY	INVENTORY METADATA
Canonical, durable business truth.	Derived current status built deterministically from events.	Raw, high-frequency metrics not suited for business event logs.	Declared, slow-changing facts about the environment.
Examples: Workflow, Run, Builder.  	Examples: Builder availability, Queue depth.  	Examples: GPU utilization, memory headroom.  	Examples: Hostnames, GPU models, tool endpoints.  

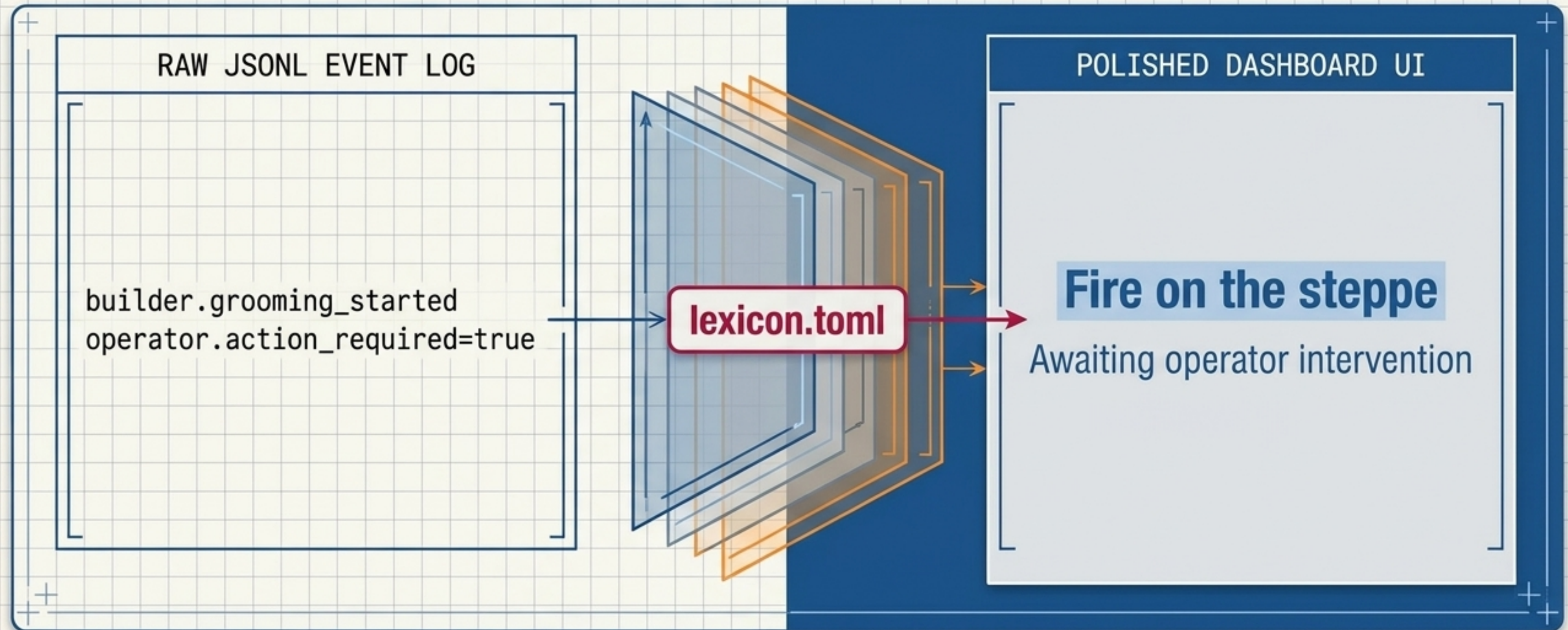
ADAPTIVE TELEMETRY VIA OPENTELEMETRY

AGENT ECOSYSTEM – OBSERVABILITY FLOW



THE LORE LAYER: BRIDGING MACHINE LOGS AND HUMAN UI

AGENT ECOSYSTEM – RENDERING ABSTRACTION

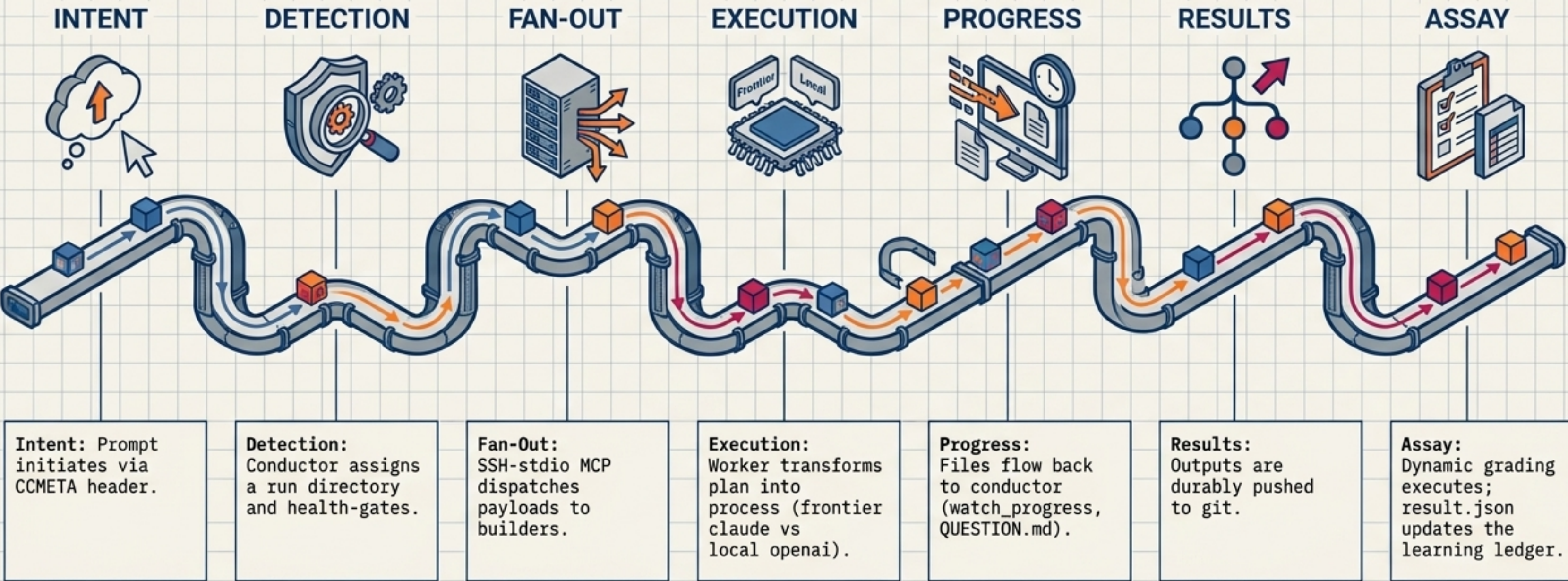


Render-time only mapping: Never write display text into the durable event ledger.

PROJECT:	
AGENT ECOSYSTEM	
DRAWING NO.:	REV.:
AE-010	1.0

SYSTEM SYNTHESIS: THE PIPELINE IN ACTION

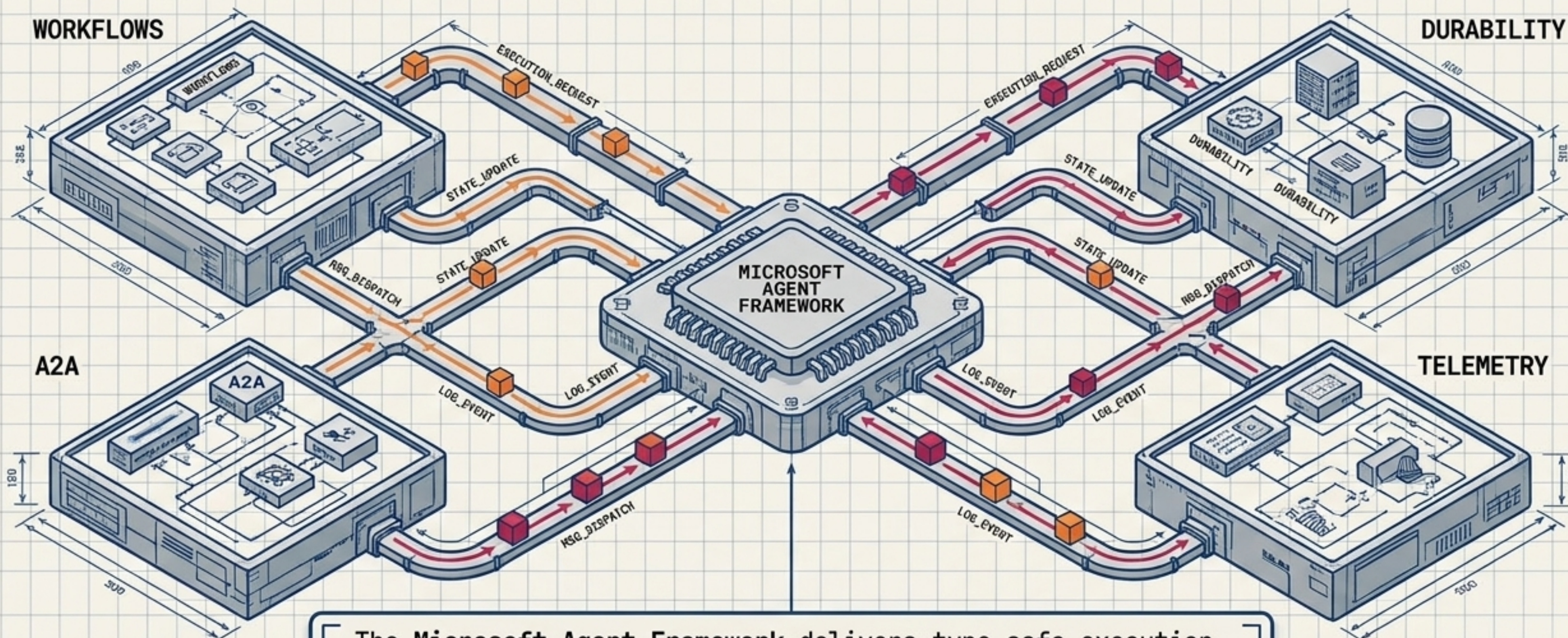
AGENT ECOSYSTEM – OBSERVABILITY FLOW



PROJECT: AGENT ECOSYSTEM	
DRAWING NO.: AE-011	REV.: 1.0

A COHESIVE NERVOUS SYSTEM FOR AI ORCHESTRATION

AGENT ECOSYSTEM – SYSTEM INTEGRATION



The Microsoft Agent Framework delivers type-safe execution, durable state resilience, seamless interoperability, and rigorous observability—transforming isolated agents into enterprise-grade operational engines.

PROJECT: AGENT ECOSYSTEM	
DRAWING NO.: AE-012	REV.: 1.0